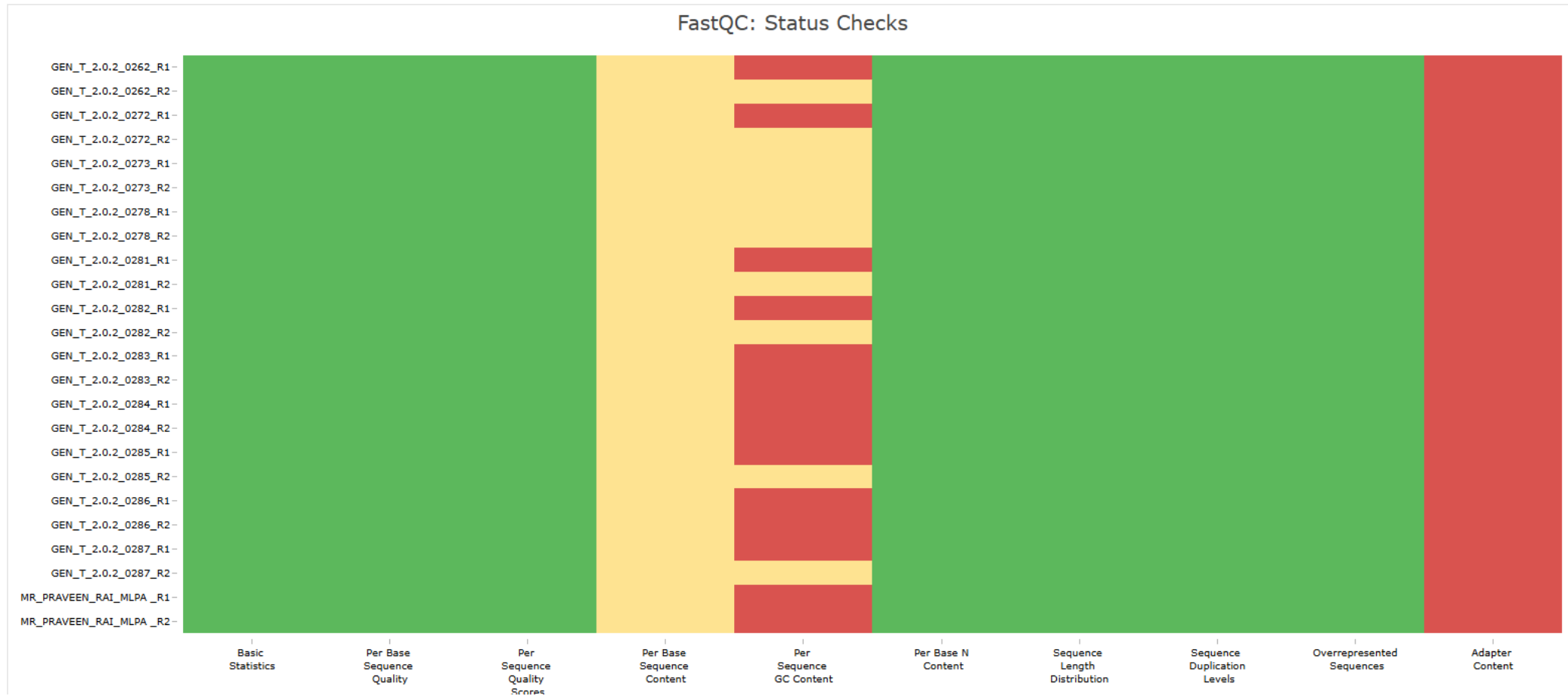


Run 23A Consolidation

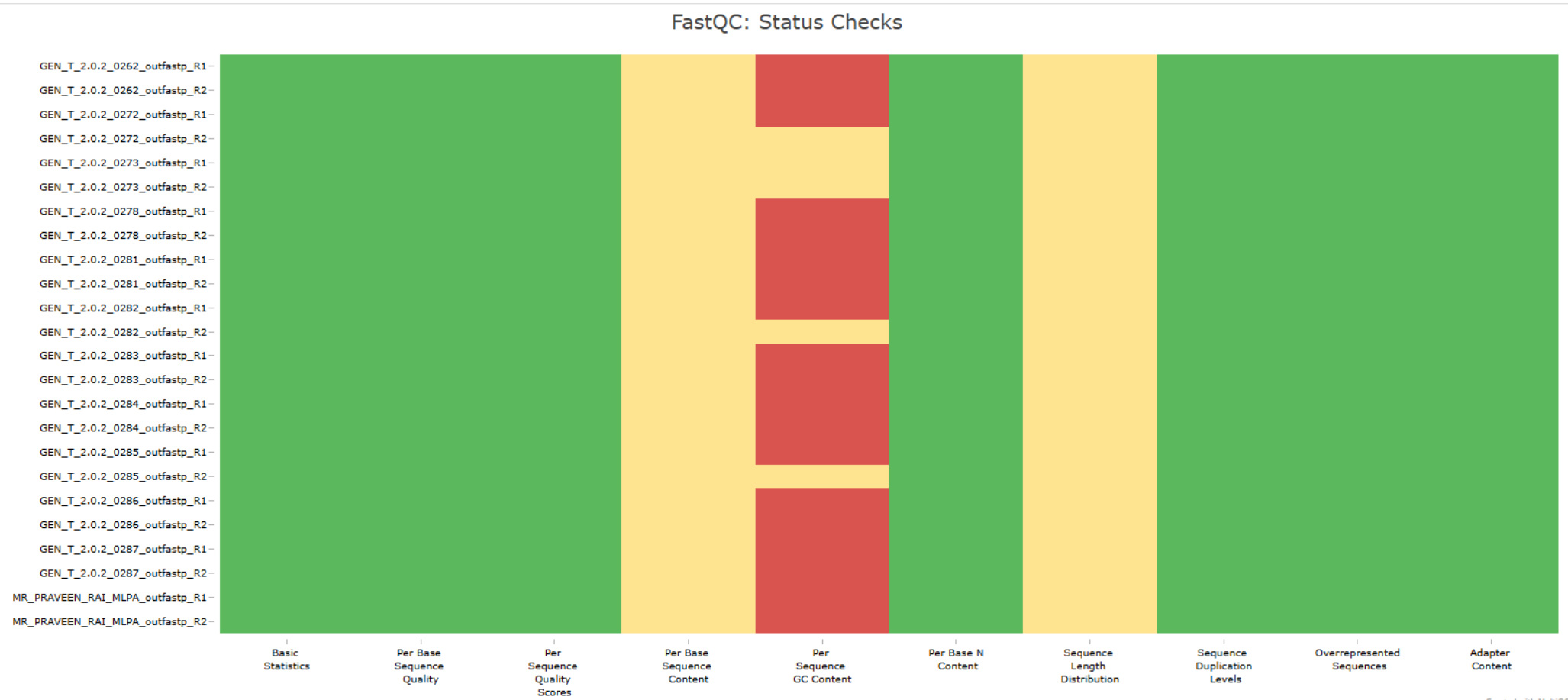
# Gender Info

| Sample_ID   | Sample_Name                                            | Gender in Freedom | Gender in IGV |
|-------------|--------------------------------------------------------|-------------------|---------------|
| TP-45-LIB01 | MRS. IRFATH SABREEN ( 31Y F ) - POC                    | Male              | Male          |
| TP-45-LIB02 | MRS. .B. SRI VIDYA W/O. DR. J. SWAMY ( 28Y F ) - POC   | Male              | Male          |
| TP-45-LIB03 | MRS.PRIYA T (22Y F) – POC                              | Female            | Female        |
| TP-45-LIB04 | MRS.UMA HANUMANTH PATIL (30Y F) - POC                  | Male              | Male          |
| TP-45-LIB05 | MRS. LIKITHA S(13959) ( 26Y F )                        | Female            | Female        |
| TP-45-LIB06 | BABY. HAMZA ( 8M M )                                   | Male              | Male          |
| TP-45-LIB07 | MRS. POOJA W/O SHARANABASAPPA ( 29Y F )                | Female            | Female        |
| TP-45-LIB08 | MR.SHARANABASAPPA (37Y M)                              | Male              | Male          |
| TP-46-LIB09 | MR. PRAVEEN RAI ( 36Y M )                              | Male              | Male          |
| TP-46-LIB10 | MR. MANIKANDAN C ( 41Y M )                             | Male              | Male          |
| TP-46-LIB11 | MRS. AYESHA SIDDIQU - (HEG M021188) ( 29Y F )          | Female            | Female        |
| TP-46-LIB12 | MR. SALAM BIN KUSSAIN SIDDIQ - (HEG M021188) ( 36Y M ) | Male              | Male          |

# Before Fastp Multiqc



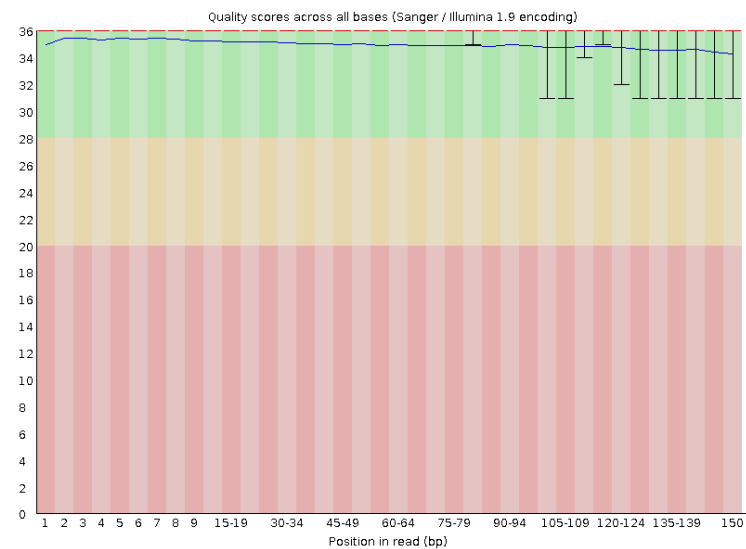
# After Fastp Multiqc



# TP-45-LIB01 Before and After Fastp

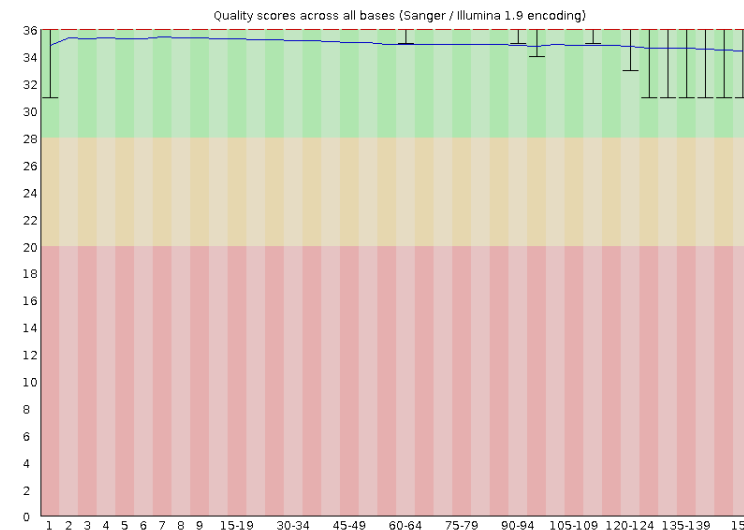
Before:  **Per base sequence quality**

R1:



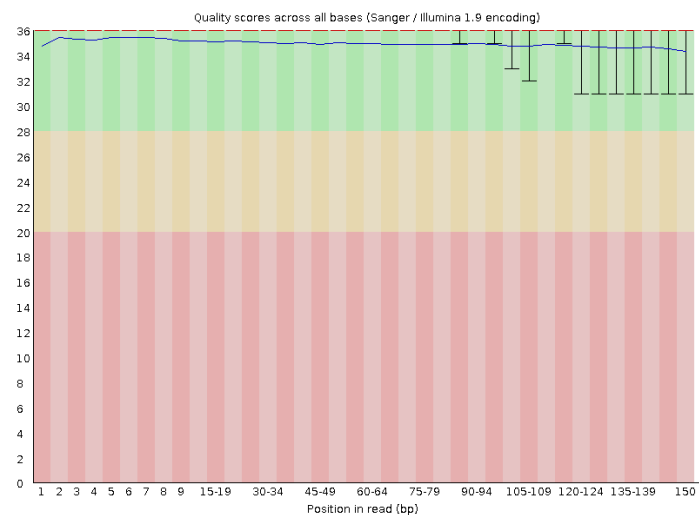
R2:

 **Per base sequence quality**



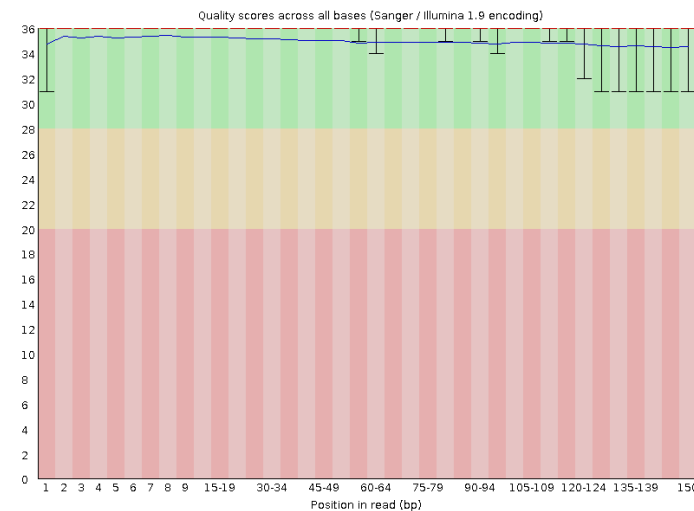
After:  **Per base sequence quality**

R1:



R2:

 **Per base sequence quality**

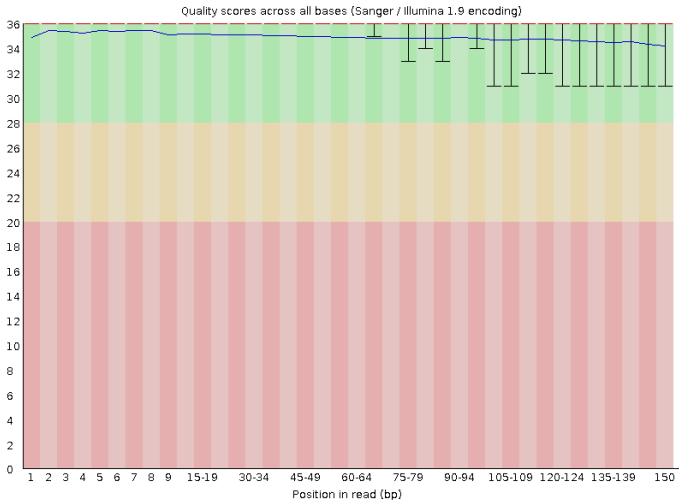


# TP-45-LIB04 Before and After Fastp

Before:

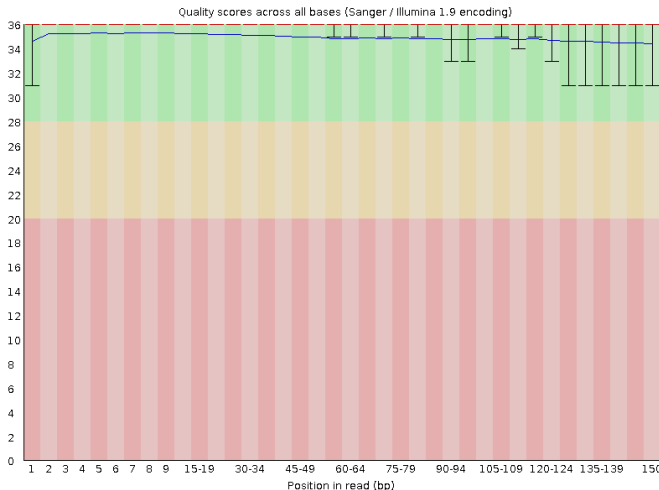
R1:

✔ Per base sequence quality



R2:

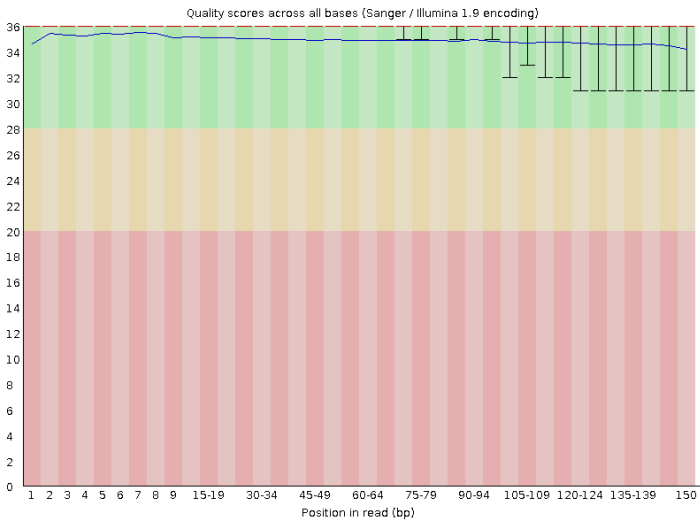
✔ Per base sequence quality



After:

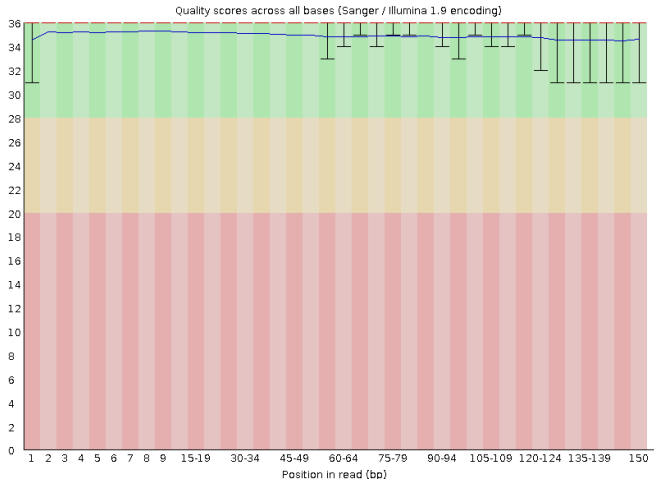
R1:

✔ Per base sequence quality



R2:

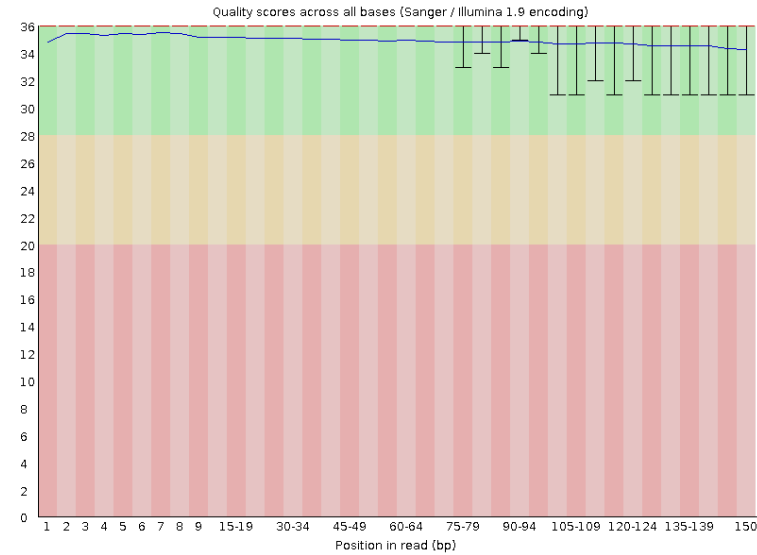
✔ Per base sequence quality



# TP-46-LIB09 Before and After Fastp

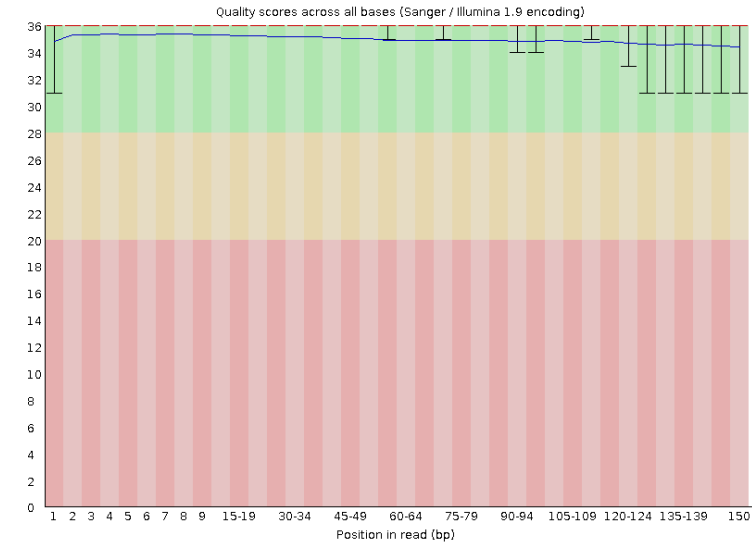
Before:  **Per base sequence quality**

R1:



R2:

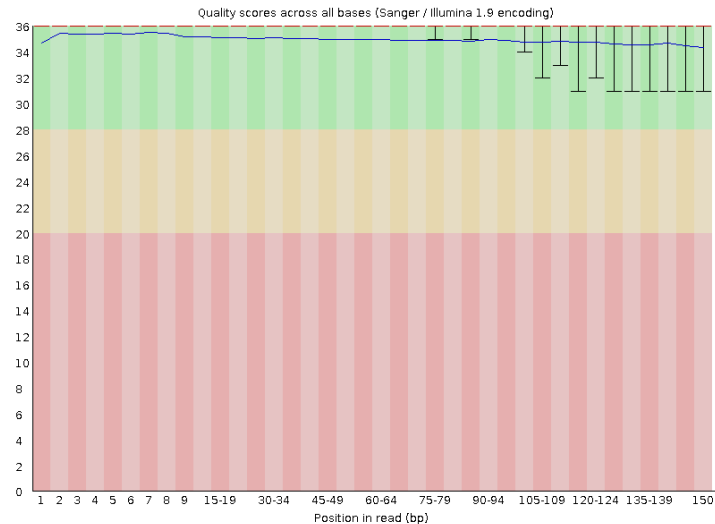
 **Per base sequence quality**



After:

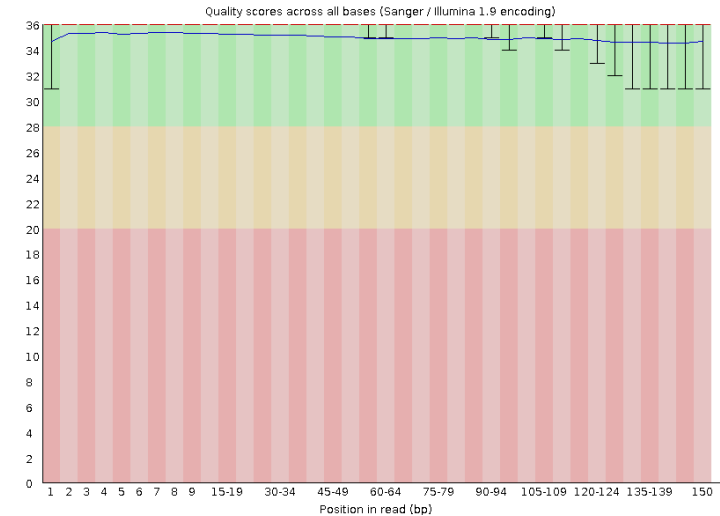
R1:

 **Per base sequence quality**



R2:

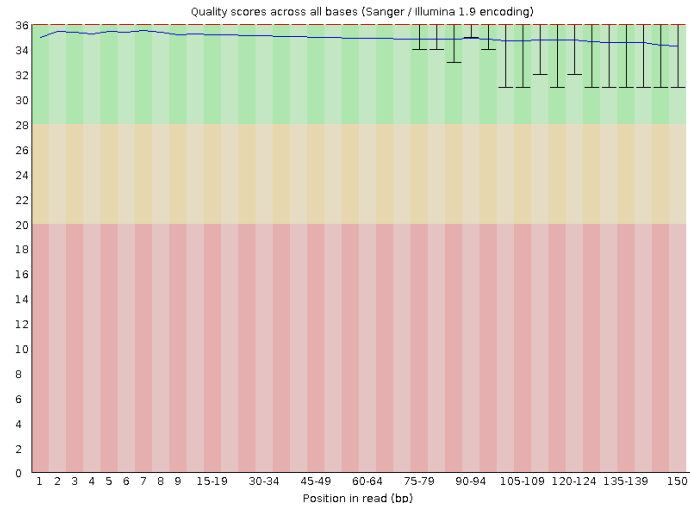
 **Per base sequence quality**



# TP-46-LIB12 Before and After Fastp

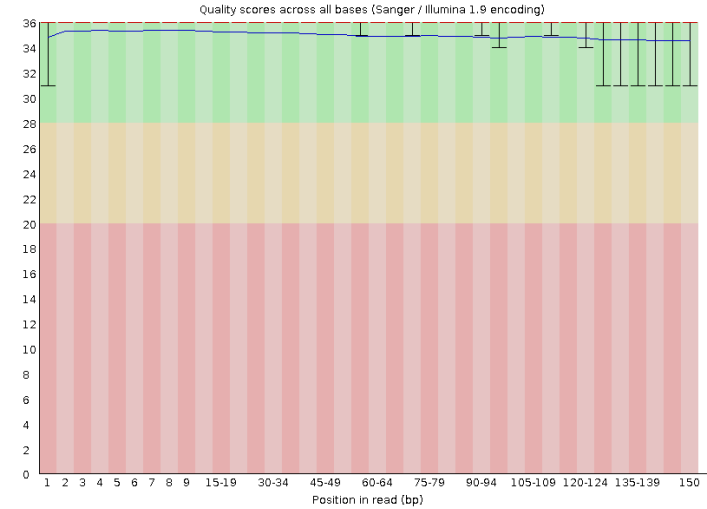
Before:  **Per base sequence quality**

R1:



R2:

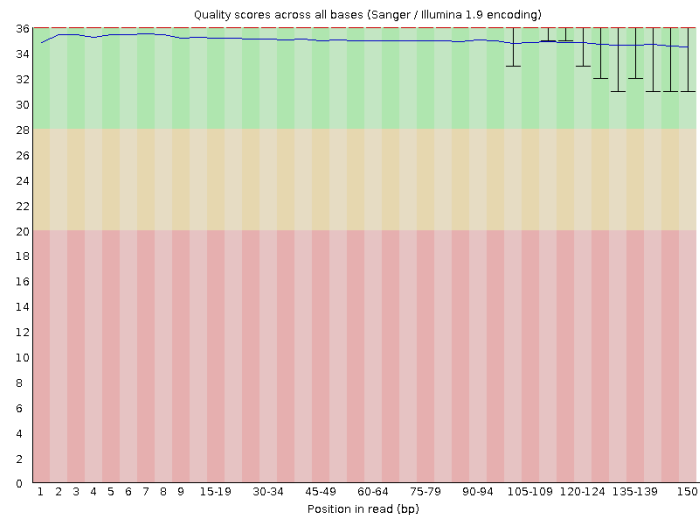
 **Per base sequence quality**



After:

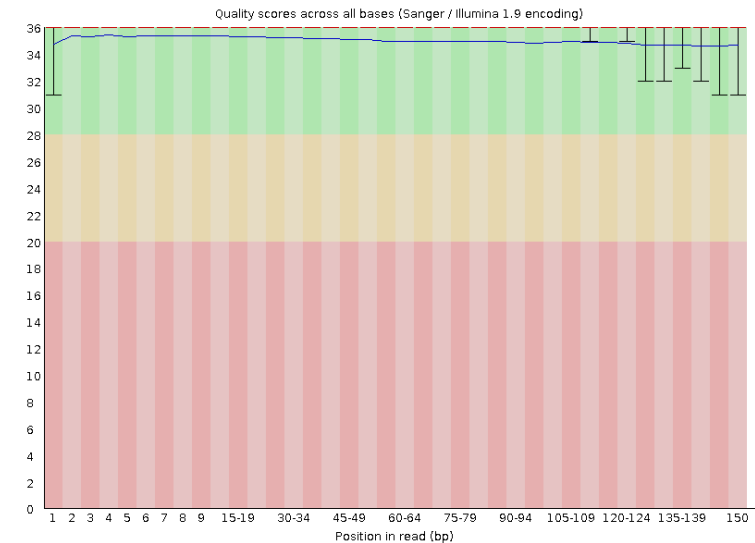
 **Per base sequence quality**

R1:



R2:

 **Per base sequence quality**





# Fastp results

| FASTP COMPARISSION RUN 23A |                                                        |                     |  |                 |               |            |                      |                 |               |            |
|----------------------------|--------------------------------------------------------|---------------------|--|-----------------|---------------|------------|----------------------|-----------------|---------------|------------|
| Sample_ID                  |                                                        | Sample_Name         |  | BEFORE FASTP    |               |            | AFTER FASTP          |                 |               |            |
|                            |                                                        |                     |  | NUMBER OF READS | SIZE OF READS | Q30        | TOTAL DATA GENERATED | NUMBER OF READS | SIZE OF READS | Q30        |
| TP-45-LIB01                | MRS. IRFATH SABREEN ( 31Y F ) - POC                    | GEN_T_2.0.2_0262    |  | 108.426580 M    | 150bp         | 95.016579% | 16.263987 G          | 108.135632 M    | 140bp         | 95.180486% |
| TP-45-LIB02                | MRS. .B. SRI VIDYA W/O. DR. J. SWAMY ( 28Y F ) - POC   | GEN_T_2.0.2_0273    |  | 123.733018 M    | 150bp         | 94.951609% | 18.559953 G          | 123.380300 M    | 142bp         | 95.088418% |
| TP-45-LIB03                | MRS.PRIYA T (22Y F) – POC                              | GEN_T_2.0.2_0278    |  | 113.713100 M    | 150bp         | 94.890406% | 17.056965 G          | 113.396396 M    | 139bp         | 95.052788% |
| TP-45-LIB04                | MRS.UMA HANUMANTH PATIL (30Y F) - POC                  | GEN_T_2.0.2_0272    |  | 111.713074 M    | 150bp         | 94.831098% | 16.756961 G          | 111.426564 M    | 140bp         | 94.979148% |
| TP-45-LIB05                | MRS. LIKITHA S(13959) ( 26Y F )                        | GEN_T_2.0.2_0281    |  | 109.207770 M    | 150bp         | 95.269414% | 16.381165 G          | 108.920834 M    | 140bp         | 95.414901% |
| TP-45-LIB06                | BABY. HAMZA ( 8M M )                                   | GEN_T_2.0.2_0282    |  | 110.868936 M    | 150bp         | 95.200666% | 16.630340 G          | 110.566850 M    | 139bp         | 95.396962% |
| TP-45-LIB07                | MRS. POOJA W/O SHARANABASAPPA ( 29Y F )                | GEN_T_2.0.2_0283    |  | 118.734742 M    | 150bp         | 94.815976% | 17.810211 G          | 118.421872 M    | 140bp         | 94.958578% |
| TP-45-LIB08                | MR.SHARANABASAPPA (37Y M)                              | GEN_T_2.0.2_0284    |  | 110.529288 M    | 150bp         | 95.133460% | 16.579393 G          | 110.228120 M    | 143bp         | 95.275149% |
| TP-46-LIB09                | MR. PRAVEEN RAI ( 36Y M )                              | MR_PRAVEEN_RAI_MLPA |  | 92.138350 M     | 150bp         | 95.007656% | 13.820753 G          | 91.887310 M     | 140bp         | 95.176599% |
| TP-46-LIB10                | MR. MANIKANDAN C ( 41Y M )                             | GEN_T_2.0.2_0285    |  | 83.156888 M     | 150bp         | 94.854589% | 12.473533 G          | 82.925320 M     | 143bp         | 94.988937% |
| TP-46-LIB11                | MRS. AYESHA SIDDIQU - (HEG M021188) ( 29Y F )          | GEN_T_2.0.2_0286    |  | 93.020910 M     | 150bp         | 94.793416% | 13.953137 G          | 92.765214 M     | 142bp         | 94.933786% |
| TP-46-LIB12                | MR. SALAM BIN KUSSAIN SIDDIQ - (HEG M021188) ( 36Y M ) | GEN_T_2.0.2_0287    |  | 84.341200 M     | 150bp         | 95.033230% | 12.651180 G          | 84.116820 M     | 142bp         | 95.184561% |

# HS metrics results

| MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000 |             |             |             |             |             |             |             |             |             |             |             |             |
|-------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| BAIT_SET                                        |             | BAIT        |             |             |             |             |             |             |             |             |             |             |
|                                                 | RUN = 23A   |             |             |             |             |             |             |             |             |             |             |             |
|                                                 | TP-45-LIB01 | TP-45-LIB02 | TP-45-LIB03 | TP-45-LIB04 | TP-45-LIB05 | TP-45-LIB06 | TP-45-LIB07 | TP-45-LIB08 | TP-46-LIB09 | TP-46-LIB10 | TP-46-LIB11 | TP-46-LIB12 |
| GENOME_SIZE                                     | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  | 3217346917  |
| BAIT_TERRITORY                                  | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    | 43249815    |
| TARGET_TERRITORY                                | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    | 36458262    |
| BAIT_DESIGN_EFFICIENCY                          | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    | 0.842969    |
| TOTAL_READS                                     | 108135632   | 123380300   | 113396396   | 111426564   | 108920834   | 110566850   | 118421872   | 110228120   | 91887310    | 82925320    | 92765214    | 84116820    |
| PF_READS                                        | 108135632   | 123380300   | 113396396   | 111426564   | 108920834   | 110566850   | 118421872   | 110228120   | 91887310    | 82925320    | 92765214    | 84116820    |
| PF_UNIQUE_READS                                 | 108135632   | 123380300   | 113396396   | 111426564   | 108920834   | 110566850   | 118421872   | 110228120   | 91887310    | 82925320    | 92765214    | 84116820    |
| PCT_PF_READS                                    | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           |
| PCT_PF_UQ_READS                                 | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           | 1           |
| PF_UQ_READS_ALIGNED                             | 108129470   | 123373790   | 113387373   | 111420073   | 108915564   | 110559420   | 118415563   | 110222529   | 91882212    | 82921407    | 92760488    | 84112988    |
| PCT_PF_UQ_READS_ALIGNED                         | 0.999943    | 0.999947    | 0.99992     | 0.999942    | 0.999952    | 0.999933    | 0.999947    | 0.999949    | 0.999945    | 0.999953    | 0.999949    | 0.999954    |
| PF_BASES_ALIGNED                                | 15225924081 | 17541014759 | 15869172721 | 15630985389 | 15278182182 | 15433524066 | 16697054220 | 15820010324 | 12898284759 | 11895562686 | 13269882499 | 12028953663 |
| PF_UQ_BASES_ALIGNED                             | 15225924081 | 17541014759 | 15869172721 | 15630985389 | 15278182182 | 15433524066 | 16697054220 | 15820010324 | 12898284759 | 11895562686 | 13269882499 | 12028953663 |
| ON_BAIT_BASES                                   | 9359646631  | 10748084637 | 9916290176  | 9693243364  | 9577526910  | 9691647228  | 10368343250 | 9543539881  | 7901288145  | 7058860854  | 7936815781  | 7186455258  |
| NEAR_BAIT_BASES                                 | 3754317633  | 4579460633  | 3943057996  | 3957096968  | 3781227696  | 3757438778  | 4201062404  | 4192263316  | 3166982539  | 3153353030  | 3478269862  | 3150636899  |
| OFF_BAIT_BASES                                  | 2111959817  | 2213469489  | 2009824549  | 1980645057  | 1919427576  | 1984438060  | 2127648566  | 2084207127  | 1830014075  | 1683348802  | 1854796856  | 1691861506  |
| ON_TARGET_BASES                                 | 5968559016  | 6800098811  | 6353044389  | 6226248076  | 6094605517  | 6200986018  | 6568017012  | 5863929442  | 5028173679  | 4411081348  | 4962234831  | 4500111249  |
| PCT_SELECTED_BASES                              | 0.861292    | 0.873812    | 0.87335     | 0.873287    | 0.874368    | 0.87142     | 0.872573    | 0.868255    | 0.85812     | 0.858489    | 0.860225    | 0.859351    |
| PCT_OFF_BAIT                                    | 0.138708    | 0.126188    | 0.12665     | 0.126713    | 0.125632    | 0.12858     | 0.127427    | 0.131745    | 0.14188     | 0.141511    | 0.139775    | 0.140649    |
| ON_BAIT_VS_SELECTED                             | 0.713716    | 0.701227    | 0.715495    | 0.71011     | 0.716948    | 0.720618    | 0.711652    | 0.694793    | 0.713868    | 0.691217    | 0.695292    | 0.695211    |
| MEAN_BAIT_COVERAGE                              | 216.409     | 248.512     | 229.279     | 224.122     | 221.447     | 224.085     | 239.732     | 220.661     | 182.69      | 163.211     | 183.511     | 166.162     |
| MEAN_TARGET_COVERAGE                            | 163.709     | 186.517     | 174.255     | 170.777     | 167.167     | 170.085     | 180.152     | 160.84      | 137.916     | 120.99      | 136.107     | 123.432     |
| PCT_USABLE_BASES_ON_BAIT                        | 0.615095    | 0.612317    | 0.62487     | 0.620492    | 0.626889    | 0.627707    | 0.621074    | 0.602924    | 0.612769    | 0.59376     | 0.598409    | 0.597682    |
| PCT_USABLE_BASES_ON_TARGET                      | 0.392241    | 0.387401    | 0.400334    | 0.39856     | 0.398917    | 0.401624    | 0.393431    | 0.370461    | 0.38995     | 0.37104     | 0.374136    | 0.374264    |
| FOLD_ENRICHMENT                                 | 45.7288     | 45.5816     | 46.4845     | 46.1314     | 46.6332     | 46.7139     | 46.1937     | 44.8762     | 45.5701     | 44.1431     | 44.4931     | 44.4427     |
| ZERO_CVG_TARGETS_PCT                            | 0.00730783  | 0.00704017  | 0.00812138  | 0.00723387  | 0.00830099  | 0.00747336  | 0.00799811  | 0.00732192  | 0.00716343  | 0.00722683  | 0.0078678   | 0.00724444  |
| PCT_EXC_DUPE                                    | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| PCT_EXC_MAPQ                                    | 0.0592333   | 0.0510264   | 0.0510193   | 0.0506713   | 0.050888    | 0.0548728   | 0.0521444   | 0.0531577   | 0.0567616   | 0.0512923   | 0.0517784   | 0.052016    |
| PCT_EXC_BASEQ                                   | 0.00569158  | 0.0057539   | 0.00591939  | 0.00602935  | 0.00539838  | 0.0053363   | 0.00596614  | 0.0053318   | 0.00560636  | 0.00570579  | 0.00577913  | 0.00537206  |
| PCT_EXC_OVERLAP                                 | 0.35484     | 0.338794    | 0.360237    | 0.350763    | 0.364317    | 0.367828    | 0.356059    | 0.331787    | 0.359582    | 0.328489    | 0.333508    | 0.334042    |
| PCT_EXC_OFF_TARGET                              | 0.723632    | 0.729038    | 0.73406     | 0.737362    | 0.728616    | 0.728684    | 0.726984    | 0.710687    | 0.726457    | 0.724267    | 0.723109    | 0.723641    |
| FOLD_80_BASE_PENALTY                            | 1.41129     | 1.35158     | 1.39404     | 1.48502     | 1.39306     | 1.40566     | 1.36479     | 1.3986      | 1.40731     | 1.40686     | 1.37482     | 1.40263     |
| PCT_TARGET_BASES_0X                             | 1.0092      | 0.9622      | 1.1134      | 1.0054      | 1.1327      | 1.0119      | 1.098       | 1.0043      | 1.0159      | 0.994       | 1.0949      | 0.9955      |
| PCT_TARGET_BASES_1X                             | 0.989908    | 0.990378    | 0.988866    | 0.989946    | 0.988673    | 0.989881    | 0.98902     | 0.989957    | 0.989841    | 0.99006     | 0.989051    | 0.990045    |
| PCT_TARGET_BASES_2X                             | 0.989241    | 0.98975     | 0.988193    | 0.989312    | 0.988017    | 0.989227    | 0.988446    | 0.989288    | 0.98921     | 0.989338    | 0.988346    | 0.98939     |
| PCT_TARGET_BASES_10X                            | 0.986756    | 0.987378    | 0.985705    | 0.986843    | 0.985666    | 0.986854    | 0.986143    | 0.986792    | 0.986598    | 0.986417    | 0.9857      | 0.986513    |
| PCT_TARGET_BASES_20X                            | 0.98461     | 0.985486    | 0.983588    | 0.984629    | 0.98366     | 0.984948    | 0.984232    | 0.984668    | 0.984143    | 0.983412    | 0.983184    | 0.983692    |
| PCT_TARGET_BASES_30X                            | 0.982269    | 0.983682    | 0.981498    | 0.982312    | 0.981656    | 0.982892    | 0.982391    | 0.982423    | 0.981181    | 0.979372    | 0.980355    | 0.979854    |
| PCT_TARGET_BASES_40X                            | 0.978976    | 0.981661    | 0.979043    | 0.979143    | 0.979188    | 0.980147    | 0.980304    | 0.979347    | 0.976352    | 0.971885    | 0.976524    | 0.973043    |
| PCT_TARGET_BASES_50X                            | 0.97361     | 0.978944    | 0.975806    | 0.974188    | 0.975697    | 0.975832    | 0.977642    | 0.974477    | 0.967704    | 0.958207    | 0.970283    | 0.960481    |
| PCT_TARGET_BASES_100X                           | 0.86949     | 0.928762    | 0.90538     | 0.869538    | 0.890474    | 0.888355    | 0.923424    | 0.872364    | 0.78641     | 0.683027    | 0.791288    | 0.700759    |
| HS_LIBRARY_SIZE                                 | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_10X                                  | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_20X                                  | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_30X                                  | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_40X                                  | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_50X                                  | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| HS_PENALTY_100X                                 | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
| AT_DROPOUT                                      | 0.0456057   | 0.0586077   | 0.0423798   | 0.0393765   | 0.0403107   | 0.0424944   | 0.0411449   | 0.0354399   | 0.0335685   | 0.0301133   | 0.0314981   | 0.0307058   |
| GC_DROPOUT                                      | 6.50434     | 4.6256      | 7.86541     | 8.93915     | 7.91825     | 7.19481     | 7.1742      | 7.52676     | 6.9734      | 7.73142     | 7.62117     | 7.60186     |
| DATA_GENERATED                                  | 15.216576 G | 17.553141 G | 15.869358 G | 15.621861 G | 15.277879 G | 15.439766 G | 16.694221 G | 15.828754 G | 12.894403 G | 11.888416 G | 13.263192 G | 12.023887 G |